

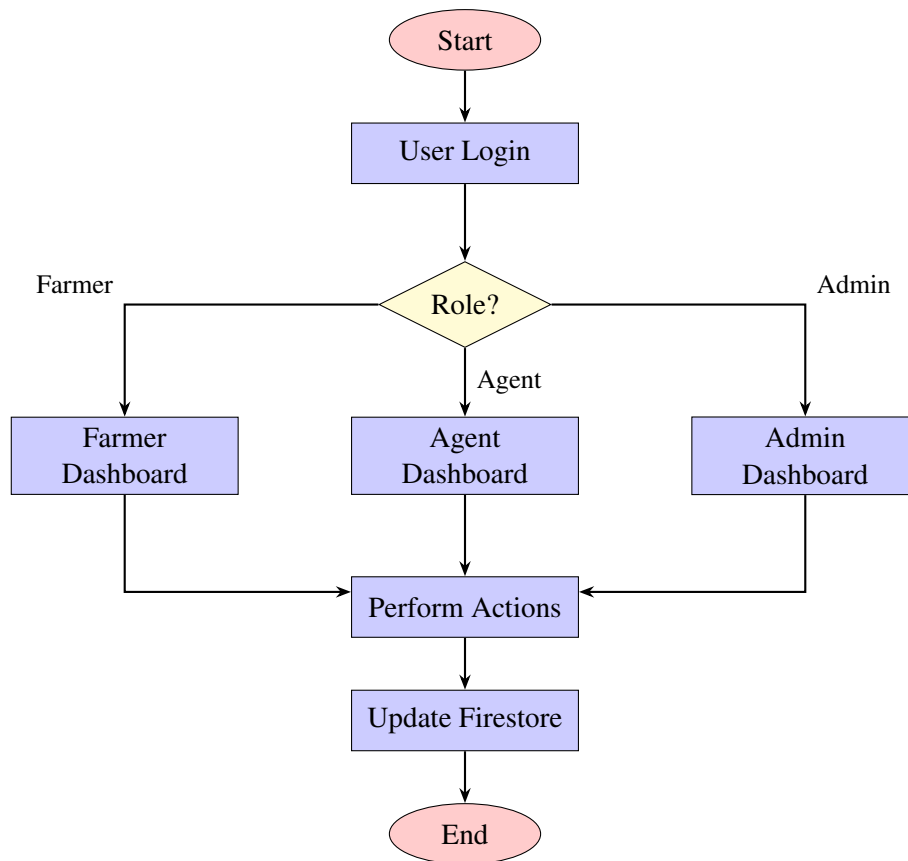


Figure 7.3: Application Flow Diagram

7.3 Flow Diagram

7.4 Use Case Diagram

7.5 Activity Diagram

7.6 ER Diagram

The ER diagram illustrates the six core entities of the AgriTrace system and their relationships. The User entity is central, connecting to Listings (creation relationship), CarbonCredits (earning relationship), and Notifications (receiving relationship). Listings generate Orders, which in turn require Payments. These relationships are implemented as Firestore collections with document references.

7.7 Firestore Collection Schema

The following table details the Firestore collections, their fields, data types, and descriptions: